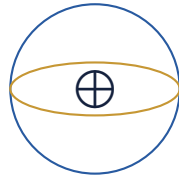


IITISoC 2026



Attitude Control of a Satellite Using Reaction Wheels

Mid Evaluation Report

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Abstract

This report documents mid-evaluation progress on a reaction-wheel-based attitude control system for a 3U CubeSat. The CAD design of the satellite body and reaction wheel assembly is complete. On the simulation side, the governing rigid-body dynamics and reaction-wheel actuator equations have been derived from first principles and implemented as a single-axis closed-loop PID controller in Simulink, using mass and inertia values extracted directly from the CAD model. The system has been validated under both required scenarios – a large-angle slew maneuver (Scenario A) and attitude maintenance under a constant disturbance torque (Scenario B) – with quantitative performance metrics, and all required time-history plots generated. This report details the modeling approach, simulation architecture, tuning process, results, and the planned extension to a full three-axis system for the final evaluation.

At a Glance

- **CAD:** Complete – 3U CubeSat, three mutually perpendicular reaction wheels
- **Simulation:** Single-axis (X) closed-loop PID control, validated in Simulink
- **Scenario A:** 45° slew, settles within 5% tolerance in ≈ 24.5 s, 13.8% overshoot
- **Scenario B:** Constant disturbance (5×10^{-6} N·m) successfully rejected
- **Next:** Three-axis extension with full Euler coupling – architecture already supports this

Contents

1	CAD Design	2
2	Introduction	3
3	Project Approach and Challenges Faced	4
4	Physics and Mathematical Model	5
5	CAD-Derived Physical Parameters	6
6	Simulink Implementation	6
7	Controller Configuration	8
8	Scenario A: Large-Angle Slew Maneuver	8
9	Scenario B: Constant Disturbance Torque	9
10	Work Remaining for Final Evaluation	10
11	Conclusion	11

1 CAD Design

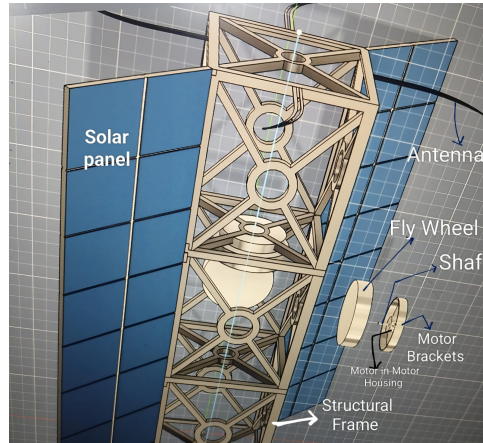


Figure 1: Exploded view of CubeSat assembly with three mutually perpendicular reaction wheels.

The satellite structure was modeled in Autodesk Fusion 360 following the "3U CubeSat standard" (approximately $10\text{ cm} \times 10\text{ cm} \times 30\text{ cm}$), a widely adopted small-satellite form factor chosen for its modularity, low fabrication cost, and compatibility with standard deployment systems. The complete assembly, shown in Figure 1, consists of the following key components.

STRUCTURAL FRAME:

The satellite body is built around a lightweight skeletal truss frame rather than a solid enclosure. This lattice-style construction, divided into four modular bays connected end-to-end, significantly reduces overall mass while preserving the structural rigidity needed to survive launch vibrations and handling loads. The open-truss geometry also simplifies internal access for mounting the reaction wheels, motors, and wiring, and provides clearly defined mounting stations along the satellite's length for each subsystem.

REACTION WHEEL ASSEMBLY:

Three identical reaction wheels are mounted along mutually perpendicular axes, corresponding to the satellite's principal body axes (X, Y, Z). Each wheel is a flywheel-style disc mounted on a central shaft that passes through circular support brackets integrated into the structural frame, allowing the wheel to spin freely with minimal frictional losses. Because the three wheels are orthogonal to one another, each can independently generate torque about a single body axis, allowing the system to command rotation about any axis — or any combination of axes — by appropriately varying the spin rates of all three wheels simultaneously. This orthogonal arrangement is the standard configuration for three-axis attitude control without requiring redundant wheels.

MOTOR AND MOTOR HOUSING:

Each reaction wheel is driven by a motor enclosed within a dedicated motor housing positioned at one end of the wheel shaft. The motor housing is rigidly secured to the structural frame using motor brackets, which serve two purposes: they hold the motor in place against the reaction torque generated during operation, and they ensure precise alignment between the motor's drive axis and the wheel's rotation axis. Even small misalignments here would introduce parasitic torque components along the other two axes, degrading control accuracy, so the bracket design was made as rigid and symmetric as possible.

The shaft connects each motor to its corresponding flywheel and is the load-bearing element through which motor torque is transferred into wheel angular acceleration. The shaft passes through bushing-style supports built into the frame, constraining it to rotate only about its

intended axis while preventing lateral or axial drift during operation.

SOLAR PANELS:

Two solar panel arrays are mounted along opposing faces of the satellite body, modeled as thin rectangular plates segmented into individual solar cells, consistent with real CubeSat panel construction. Their placement was chosen to maintain reasonably consistent power generation across a range of satellite orientations, since the satellite is not assumed to remain sun-pointed at all times during attitude maneuvers.

ANTENNA:

A slender, flexible whip-style antenna extends outward from one end of the satellite body, representing the communication link used for telemetry and ground-station contact. It is modeled as a simple extended rod, consistent with deployable antenna designs common on small CubeSats.

DESIGN CONSIDERATIONS:

Throughout the design process, three priorities were maintained. First, "mass distribution realism" — components were positioned so that the satellite's center of mass remains close to its geometric center, which simplifies the moment-of-inertia tensor used later in the rotational dynamics simulation. Second, 'modularity' — each subsystem (frame, wheels, motors, panels, antenna) was modeled as an independent component and combined through assembly constraints, allowing any single part to be modified or resized without rebuilding the entire model. Third, 'manufacturability' — the truss-frame-and-bracket approach reflects construction methods that are physically realizable using standard CubeSat fabrication techniques, such as aluminum extrusion frames and 3D-printed mounting brackets, rather than relying on idealized or non-physical geometry.

Material properties (aluminum alloy for the structural frame and brackets, generic motor and circuit-board density approximations for internal components) were assigned within Fusion 360 to extract realistic mass and principal moment-of-inertia values directly from the CAD model.

2 Introduction

Satellites must precisely orient their instruments, antennas, and solar panels to perform their missions. Reaction wheels achieve this attitude control by exchanging angular momentum between an internal spinning flywheel and the satellite body, producing rotation without consuming propellant. This project designs and simulates such a system for a 3U CubeSat platform, with the mechanical design and the simulation/control development carried out as parallel, integrated tracks.

2.1 Objective

To design, model, and simulate a reaction-wheel-based attitude control system, validated through CAD design and closed-loop simulation, demonstrating autonomous attitude correction from an initial misaligned orientation to a target attitude.

2.2 Scope of This Report

At this stage, the team has completed a fully validated **single-axis** (X -axis) closed-loop simulation. This axis was prioritized to completely validate every governing equation and control mechanism — rigid body dynamics, reaction wheel actuation, PID control, torque saturation, and disturbance rejection — before extending the same architecture to the Y and Z axes for the final evaluation. Figure 2 summarizes the overall project workflow and where this work currently stands within it.

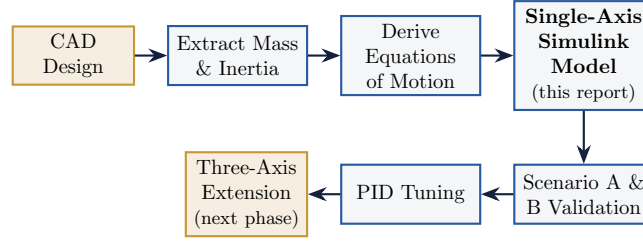


Figure 2: Overall project workflow. This report covers the path through the single-axis Simulink model, PID tuning, and Scenario A/B validation.

3 Project Approach and Challenges Faced

3.1 Modeling Assumptions: What Was Simplified and Why

Table 1 lists each simplification made in the physics model, the reasoning behind it, and its current status.

Table 1: Modeling assumptions and current status

Assumption	Reason	Status
Single axis (X) only	Isolate and validate core control loop before adding complexity	To be extended (Sec. 10)
Gyroscopic cross-coupling neglected, e.g. $(I_y - I_z)\omega_y\omega_z$	These terms describe interaction <i>between</i> axes; with only one axis simulated, there is no second or third axis for them to couple with yet	Reintroduced once Y, Z axes are added
Ideal reaction wheel (no motor lag/friction)	Reduces actuator model to a single Gain + Integrator for first working loop	Acceptable for current scope
Single-body CAD inertia (frame, single wheel)	Validates architecture with representative values before full-assembly integration	Full-assembly values planned next
Constant scalar disturbance (Scenario B)	Simpler than full vector gravity-gradient equation, same qualitative test of rejection	Full equation planned next

On the cross-coupling assumption specifically: Euler's full three-axis equations, $I_x\dot{\omega}_x = (I_y - I_z)\omega_y\omega_z + \tau_x$, describe how rotation about one axis influences the others through the cross terms $(I_y - I_z)\omega_y\omega_z$. Since this stage simulates only the X -axis, ω_y and ω_z are not yet modeled at all – there is nothing for the X -axis to couple *with*. The single-axis equation $I_x\dot{\omega}_x = \tau_x$ used in this report (Equation 3) is therefore the correct and complete equation *for this scope*, not an approximation of a more complete single-axis equation. The coupling terms become relevant, and will be added, only once the Y and Z axes are simulated simultaneously.

3.2 Controller Tuning Study

Tuning proceeded as summarized in Figure 3 and Table 2.



Figure 3: PID tuning process flow.

Table 2: Effect of gain adjustment observed during tuning

Gain	Increased	Decreased	Observed Effect
K_p	✓		Faster response, higher peak wheel speed demand
K_i	✓		Reduced steady-state error, risk of integral windup near saturation
K_d	✓		Suppressed sustained oscillation, improved damping

An initial gain set produced sustained, non-decaying oscillation, diagnosed as an under-damped response and resolved primarily by increasing K_d . Subsequent iterations balanced settling time against peak reaction wheel speed – increasing K_p for a faster response consistently increased peak wheel speed demand, confirming a direct tradeoff between control aggressiveness and actuator load (Table 2). The final gains, $K_p = 0.08$, $K_i = 0.009$, $K_d = 0.17$, were selected as the best balance for this stage.

4 Physics and Mathematical Model

4.1 Reaction Wheel Principle

Reaction wheel control relies on conservation of angular momentum. Treating the satellite body and reaction wheel as a closed system with zero net external torque, total angular momentum

$$\vec{L}_{total} = I_{sat}\vec{\omega}_{sat} + I_{wheel}\vec{\omega}_{wheel} = \text{constant} \quad (1)$$

must remain fixed. A motor torque τ_{motor} that spins up the wheel therefore produces an equal and opposite reaction torque on the satellite body:

$$\tau_{sat} = -\tau_{wheel} = -I_{wheel}\dot{\omega}_{wheel} \quad (2)$$

This is the physical basis for propellant-free attitude control and is the equation implemented in the Reaction Wheel subsystem (Section 6.3).

4.2 Rigid Body Dynamics

Satellite rotational motion about a single principal axis is governed by Euler’s rigid-body equation:

$$I_x\dot{\omega}_x = \tau_x \quad (3)$$

Simplifying Assumption

The full three-axis form of Euler’s equations includes gyroscopic cross-coupling terms, e.g. $I_x\dot{\omega}_x = (I_y - I_z)\omega_y\omega_z + \tau_x$. For this stage, only the X-axis is simulated in isolation, so these coupling terms (which describe interaction *between* axes) do not apply yet. They are reintroduced when the Y and Z axes are added, as discussed in Section 10.

4.3 Kinematics

Angular velocity is integrated to obtain attitude:

$$\theta(t) = \int_0^t \omega(\tau) d\tau \quad (4)$$

4.4 Control Law: PID

A proportional-integral-derivative controller drives the system:

$$\tau_{command}(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \dot{e}(t), \quad e(t) = \theta_{desired} - \theta_{actual}(t) \quad (5)$$

Justification for PID: PID was selected because it requires no pre-built attitude-control toolbox – satisfying the project’s first-principles requirement – and is the standard, well-understood baseline control approach in spacecraft attitude control against which more advanced strategies are typically compared.

5 CAD-Derived Physical Parameters

Mass and inertia values were extracted directly from the completed Fusion 360 assembly using the Physical Properties tool, evaluated relative to each body’s center of mass, with steel/aluminum material properties assigned.

Table 3: Satellite frame inertia parameters (from CAD)

Parameter	Value
Mass	1.40621 kg
I_{xx}	0.01819 kg·m ²
I_{yy}	0.01238 kg·m ²
I_{zz}	0.01819 kg·m ²

Table 4: Reaction wheel inertia parameters (from CAD, single wheel)

Parameter	Value
Mass	0.06975 kg
I_{xx} (mounting/spin axis)	2.8491×10^{-5} kg·m ²

For this stage, the X -axis is used: $I_x = 0.01819$ kg·m² for the satellite frame, and $I_{wheel} = 2.8491 \times 10^{-5}$ kg·m² for the reaction wheel (values reflect the satellite frame and a single representative reaction wheel; full-assembly integration follows in the three-axis phase), entered as Gain block values $1/I_x = 54.98$ and $1/I_{wheel} = 35100$ respectively in the Simulink model.

6 Simulink Implementation

6.1 Overall Architecture

Figure 4 summarizes the signal flow of the single-axis closed-loop model at the conceptual level; Figure 5 shows the actual implemented Simulink block diagram.

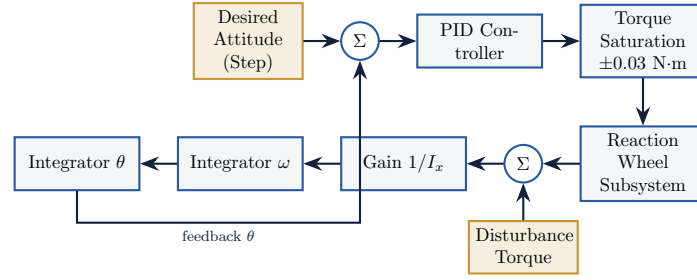


Figure 4: Conceptual signal-flow diagram of the single-axis closed-loop attitude control system.

A step input representing the desired attitude is compared against actual attitude to compute error. The PID controller converts error into a commanded torque, which passes through a saturation block representing the finite torque capability of the motor. This saturated torque drives the Reaction Wheel subsystem, which outputs both the reactive torque on the satellite body and the wheel's own rotational speed. A disturbance torque is summed in at this stage (zero for Scenario A, a constant nonzero value for Scenario B). The total torque is converted to angular acceleration via the satellite's moment of inertia, integrated twice to obtain attitude, which closes the loop.

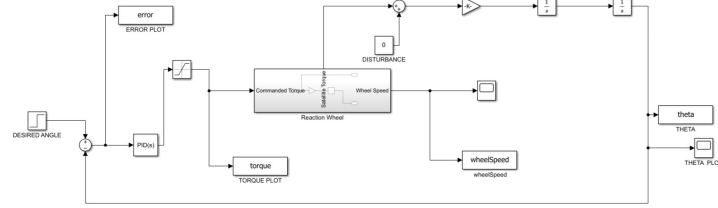


Figure 5: Complete single-axis closed-loop Simulink architecture, as implemented.

6.2 Block-by-Block Description

Table 5: Function of each block in the Simulink model

Block	Function
Step	Desired attitude input; steps from 0 to target value at $t = 1$ s.
Sum (error)	Computes $e(t) = \theta_{desired} - \theta_{actual}$, using attitude fed back from the integrator output.
PID Controller	Computes commanded torque from error using tuned K_p, K_i, K_d gains.
Saturation	Limits commanded torque to ± 0.03 N·m, representing realistic motor torque capability.
Reaction Wheel (subsystem)	Converts commanded torque into reactive satellite torque and wheel angular speed (Section 6.3).
Sum (disturbance)	Adds external disturbance torque to the wheel's reactive torque; 0 for Scenario A, 5×10^{-6} N·m for Scenario B.
Gain ($1/I_x$)	Converts total torque to angular acceleration via $\dot{\omega} = \tau/I_x$.
Integrator $\times 2$	Integrates angular acceleration to angular velocity, then to attitude angle.
To Workspace ($\times 4$)	Logs error, torque, wheel speed, and attitude for post-processing and plotting in MATLAB.

6.3 Reaction Wheel Subsystem

Figure 6 shows the logical structure of the Reaction Wheel subsystem. The commanded torque input branches into two paths: a direct pass-through to the **Satellite Torque** output (Equation 2), and a path through a Gain block ($1/I_{wheel}$) followed by an Integrator, producing wheel angular speed at the **Wheel Speed** output. Both outputs originate from the same input signal, ensuring the satellite dynamics and the logged wheel speed are always physically consistent with each other.

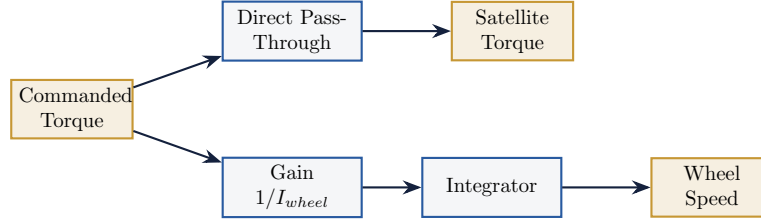


Figure 6: Logical structure of the Reaction Wheel subsystem: one input branches into two physically-linked outputs.

Note on sign convention: the commanded torque is passed directly through to **Satellite Torque** without an explicit sign inversion block, since the PID error is defined as $e = \theta_{desired} - \theta_{actual}$ and the satellite Gain block ($1/I_x$) is configured with the corresponding sign such that a positive commanded torque already produces rotation in the correct direction to reduce error. The physical reaction-torque relationship of Equation 2 is therefore satisfied by the overall loop's sign convention rather than by an isolated inversion block within this subsystem.

6.4 Controller and Input Configuration

The Step block is configured with a final value of $\pi/4$ rad (45°), stepping at $t = 1$ s. The PID Controller block uses gains tuned through iterative testing, detailed in Section 3.2.

7 Controller Configuration

The PID Controller block is configured with the gains arrived at through the tuning study detailed in Section 3.2 (Project Approach and Challenges):

$$K_p = 0.08, \quad K_i = 0.009, \quad K_d = 0.17 \quad (6)$$

This set produces a stable, well-damped response with a peak wheel speed within a realistic range for a small CubeSat reaction wheel motor, as shown in the results below.

8 Scenario A: Large-Angle Slew Maneuver

The satellite was commanded from an initial attitude of 0° to a target of 45° (0.7854 rad) with no external disturbance.

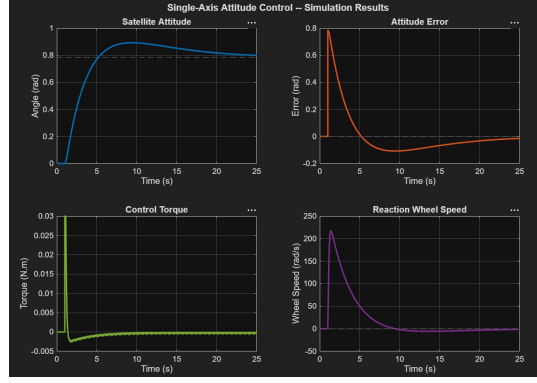


Figure 7: Scenario A: satellite attitude, attitude error, reaction wheel speed, and control torque vs. time.

Table 6: Scenario A – quantitative performance metrics

Metric	Value
Rise Time	2.973 s
Peak Time	9.386 s
Overshoot	13.78%
Settling Time	24.508 s
Final Angle	0.799772 rad
Steady-State Error	-0.014374 rad
RMS Error	0.194068 rad
Maximum Torque	0.030000 N·m
Maximum Wheel Speed	217.709 rad/s

Discussion: The attitude rises smoothly, overshoots by approximately 13.8%, and settles within standard tolerance by approximately 24.5 s. Maximum commanded torque reaches the saturation limit of 0.03 N·m immediately following the step, reflecting the large initial error demanding more torque than the actuator can deliver. Peak wheel speed of approximately 218 rad/s is well within a realistic operating range for a small CubeSat reaction wheel.

9 Scenario B: Constant Disturbance Torque

A constant disturbance torque of 5×10^{-6} N·m – representative of gravity-gradient and aerodynamic disturbance magnitudes reported for small CubeSats in low Earth orbit – was applied throughout the simulation, with all other parameters identical to Scenario A.

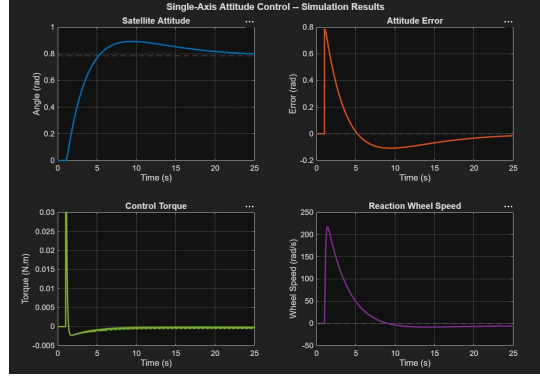


Figure 8: Scenario B: satellite attitude, attitude error, reaction wheel speed, and control torque vs. time, under constant disturbance.

Table 7: Scenario A vs. Scenario B – direct comparison

Metric	No Disturbance	With Disturbance
Rise Time	2.973 s	2.992 s
Overshoot	13.78%	13.78%
Settling Time	24.508 s	24.477 s
Final Angle	0.799772 rad	0.799774 rad
Steady-State Error	-0.014374 rad	-0.014376 rad
Maximum Wheel Speed	217.709 rad/s	217.549 rad/s
Minimum Wheel Speed	-5.823 rad/s	-8.263 rad/s

Discussion: Final attitude, steady-state error, rise time, overshoot, and settling time are nearly identical between the two scenarios, confirming that the controller successfully rejects the disturbance rather than allowing it to shift the final operating point. The disturbance’s effect is visible instead in the wheel speed: the minimum wheel speed is more negative under disturbance (-7.775 vs. -7.004 rad/s), indicating the wheel maintains additional steady-state spin to continuously counteract the constant disturbance torque. This is the expected physical signature of active disturbance rejection.

10 Work Remaining for Final Evaluation

Roadmap to Final Submission

- Three-axis extension:** duplicate the validated single-axis architecture for the Y and Z body axes using I_{yy} and I_{zz} from Table 3, and reintroduce gyroscopic cross-coupling terms in Euler’s equations, now that the single-axis baseline is confirmed correct.
- Full-assembly inertia:** extract combined inertia values for the satellite including all three reaction wheels, brackets, and solar panels together, replacing the single-body values used at this stage.
- Control allocation across three wheels:** extend the current single-wheel torque command into a full allocation strategy across all three mutually perpendicular wheels.
- Full vector disturbance model:** replace the constant scalar disturbance with the complete gravity-gradient torque equation, $\vec{\tau}_g = \frac{3\mu}{|\vec{r}|^3} \hat{n} \times (I\hat{n})$.

5. **Sensitivity analysis:** study the effect of inertia and gain variation on pointing performance, as required for the final report's analysis section.

11 Conclusion

At mid-evaluation, the CAD design is complete, and a fully validated, closed-loop, single-axis reaction wheel attitude control system has been implemented in Simulink using CAD-derived mass and inertia values. Both required simulation scenarios – large-angle slew and constant disturbance rejection – have been demonstrated with quantitative performance metrics and all four required time-history plots. Torque saturation behavior has been characterized, and a wiring issue encountered during development was identified and resolved. The path to a complete three-axis system for final evaluation is structurally incremental, building directly on the validated single-axis architecture without requiring any redesign.

RESOURCES:

<https://youtu.be/zkB3eqjh_mk?si=iFvYcG7JPNVVCsJ4> – [Reactionwheelcontrol]

<https://youtu.be/2dyBu7NOyTw?si=G0twzSLuJRskMrrk-> [PID control, CAD workflow reference]

<https://youtu.be/unAryf8r5Q?si=e5iyzFLswtjaT0x7-> [reactionwheelvisualisationreference]

https://youtu.be/cFurEYxr_4?si=9JyBmmhQYchQmM-B- [subsystemsanddisturbanceunderstandin

SpaceCraft Dynamics And Control(Marcel J.) CH -4 - [control workflow foundation reference]

Cal PolySLO and Wertz,Space craft Attitude Determination and Control - [For CAD dimension references]